

Probabilistic Machine Learning

Lecture 03: Linear Models — Classical and Probabilistic

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Module 1

Linear Regression, Classically

Guiding Question: Why squared error — and what exactly does "linear" forbid?

The Task: Predicting Acceleration

The world

O — vehicles

$Y = \mathbb{R}$ — time to accelerate to 100 km/h

$\gamma(o)$ — the true acceleration time of vehicle o

The model world

α — measure wheels, horsepower, weight, torque

X — those four numbers, for every vehicle

h_{Θ} — a linear function of them

The Data

wheels	horsepower	weight	torque	acceleration time
4	150	1350	250	9.8
4	320	1600	430	4.9
6	95	2100	210	16.2
...	...	...	...	...

The last column is y the rest is $\mathbf{x}$.

The Remaining Two Pieces

Loss function

Mean squared error — the average of the squared differences between predicted and observed acceleration times.

Learning algorithm

LMS — least mean squares, which adjusts the parameters to reduce that error.

Four choices — representation, model family, loss, optimiser — and the system is fully specified. Every method in this chapter is a different set of four.

A learning problem becomes concrete the moment you can name its **task, experience and performance measure**, and its system becomes buildable the moment you can name its **representation, model, loss and optimiser**.

Fitting h by Repeated Correction

LMS (least mean squares)

Initialise the weights w .

repeat pick a training example $(\mathbf{x}, y)$ compute the prediction $h_w(\mathbf{x})$ compute the error $e = y - h_w(\mathbf{x})$ update every weight: $w_i \leftarrow w_i + \eta e x_i$ **until** convergence on D

What the Update Rule Encodes

Direction

The sign of e decides whether the prediction goes up or down.

Attribution

The factor x_i gives each feature credit proportional to its value.

Step size

The learning rate η trades speed against stability.

Too large an η oscillates or diverges too small an η converges slowly. It is a hyperparameter — chosen, not learned.

Classification versus Regression

Classification

Y is a finite set of labels.

Will this customer default — yes or no?

Regression

$Y = \mathbb{R}$, the target is a quantity.

What credit line should this customer receive?

The learning machinery is the same only the shape of Y — and therefore the loss — changes.

The Model

Linear regression predicts the target as a weighted sum of the features:

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + \dots + w_px_p = \mathbf{w}^\top \mathbf{x}$$

with $x_0 = 1$ absorbing the intercept.

The hypothesis space is $\mathbb{R}^{p+1}$: one hypothesis per weight vector.

The Objective: Residual Sum of Squares

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathbb{R}^{p+1}} \text{RSS}(\mathbf{w}), \quad \text{RSS}(\mathbf{w}) = \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

RSS measures **effectiveness** — how good the predictions are. Do not confuse it with **efficiency**, which is about runtime and computational effort.

One Feature: The Closed Form

For a single feature the minimiser can be written down directly:

Slope

$$\hat{w}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

The covariance of x and y over the variance of x .

Intercept

$$\hat{w}_0 = \bar{y} - \hat{w}_1 \bar{x}$$

The fitted line passes through the point of means.

How Much Variance Is Explained?

$$\text{TSS} = \text{ESS} + \text{RSS}, \quad R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

where TSS is the total, ESS the explained and RSS the residual sum of squares.

R^2 is the fraction of the target's variation the model accounts for: 0 means no better than predicting the mean, 1 means an exact fit.

Training R^2 never decreases when a feature is added, however useless the feature. A high training R^2 is not evidence that a model generalises — test R^2 and adjusted R^2 can and do fall.

Higher Dimensions

With p features the same objective in matrix form,

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the optimum in one step whenever $\mathbf{X}^\top \mathbf{X}$ is invertible.

Invertibility fails when features are linearly dependent or when $p > n$ — more unknowns than equations. Both are common in practice, and both are why regularisation exists.

Takeaway

Linear regression fits a weighted sum by minimising squared residuals. Its solution is built from **covariance over variance**, and R^2 reports how much variation it explains — not whether it will generalise.

Linear in the Parameters, Not in the Features

A model is **linear** when it is linear in the **weights** $\mathbf{w}$. The features may be arbitrary transformations of the input.

Plain

$$h(\mathbf{x}) = w_0 + w_1x$$

a straight line

Polynomial

$$h(\mathbf{x}) = w_0 + w_1x + w_2x^2$$

a curve — still a *linear model*

Basis functions

$$h(\mathbf{x}) = \sum_j w_j \phi_j(\mathbf{x})$$

any fixed feature map ϕ

Consequences

The good

The closed-form solution and all statistical guarantees carry over unchanged.

The catch

The feature map is a **modelling decision**. Choose too rich a basis and the model overfits too poor and it underfits.

Increasing polynomial degree is a dial on the bias–variance axis you can watch turning.

The Question Nobody Asks

We minimised the *squared* residuals. Why squares — why not absolute deviations, or fourth powers?

Geometric View

The predictions $\mathbf{X}\mathbf{w}$ live in the **column space** of $\mathbf{X}$ — everything the features can express. The observed $\mathbf{y}$ generally does not.

Least squares picks the point of that space **closest to $\mathbf{y}$ in Euclidean distance**: the orthogonal projection. The residual is perpendicular to every feature.

Consequence

The residual carries no remaining linear information about the features.

The Gauss–Markov Theorem

Gauss–Markov (informal). Among all estimators that are

- **linear** in the observed outputs $(y_1, \dots, y_n)$, and
- **unbiased**,

the least-squares estimator has the **lowest variance**. It is the *Best Linear Unbiased Estimator* — BLUE.

Best

lowest variance among the candidates

Linear

a linear function of the observations

Unbiased

right on average over repeated samples

What This Buys, and What It Costs

The payoff

Minimising RSS yields optimal statistical properties under **very mild assumptions** — no normality needed, only uncorrelated errors of constant variance.

The price

Alternatives — absolute error, Huber loss — are **more robust to outliers** but give up the BLUE guarantee.

Squared error is not an arbitrary convenience: it is the loss whose minimiser is the best linear unbiased estimator. Departing from it is legitimate — but it is a trade, and you should be able to name what you traded.

Two Ways to the Same Minimum

Direct (closed form)

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Exact, one step. Requires forming and inverting a $(p+1) \times (p+1)$ matrix — costly for large p , impossible when it is singular.

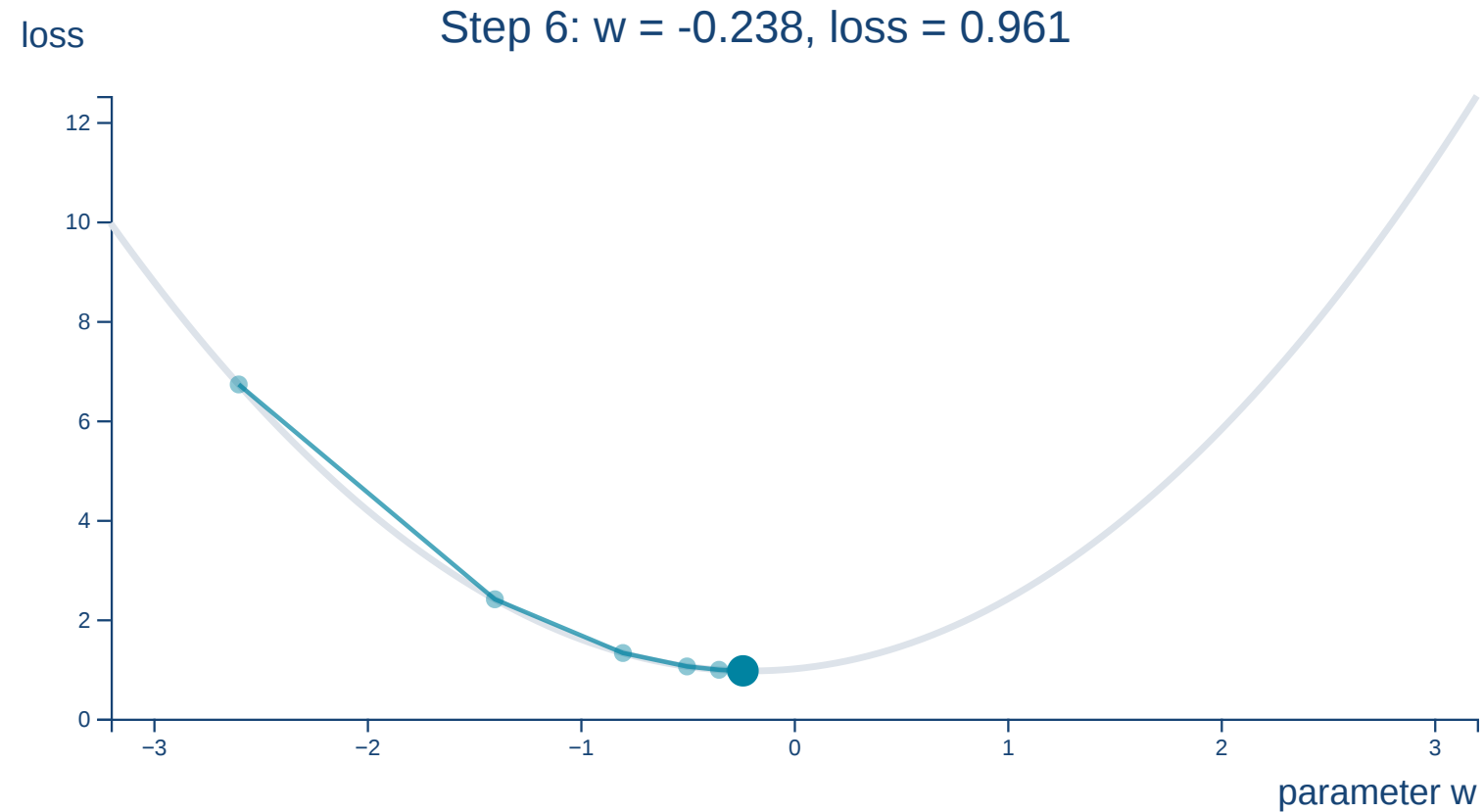
Iterative (gradient descent)

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla \text{RSS}(\mathbf{w})$$

Approximate, many steps. Scales to large p and n , works when the direct solution does not exist, and extends to models with no closed form at all.

The iterative rule is exactly the **LMS algorithm** from Module 2: the update $w_i \leftarrow w_i + \eta e x_i$ is a step along $-\nabla \text{RSS}$, written per example instead of per dataset.

Watching Gradient Descent Work



The Learning Rate Decides Everything



Teal: $\eta = 0.9$ — overshoots and oscillates across the minimum. Red: $\eta = 0.05$ — stable, but barely moves.



The learning rate is a **hyperparameter**: it is chosen, not learned, and both failure modes are silent — one looks unstable, the other looks converged.

Takeaway

Use the **closed form** when the matrix is small and well-conditioned use **gradient descent** when it is not — which, for everything after this unit, is always.

Module 2

The Probabilistic Reading

Guiding Question: *Least squares was justified after the fact. What if we had stated how the data arose first — would we get the same estimator?*

Turning a Model Into an Estimator

We have a model with parameters θ and data D . The model assigns each possible dataset a probability.

The **likelihood** is that probability, read as a function of the **parameters** with the data held fixed:

$$L(\theta) = p(D \mid \theta)$$

The **maximum likelihood estimate** is $\hat{\theta}_{\text{ML}} = \arg\max_{\theta} p(D \mid \theta)$.

$p(D \mid \theta)$ is a probability over **data**, not over parameters. The likelihood is not a distribution over θ and does not integrate to 1.

Why the Logarithm

Under the **i.i.d. assumption** — observations independent and identically distributed — the likelihood factorises:

$$L(\theta) = \prod_{i=1}^n p(x_i \mid \theta) \quad \implies \quad \ell(\theta) = \sum_{i=1}^n \log p(x_i \mid \theta)$$

Same maximum

log is monotone, so it does not move the arg max.

Numerically safe

A product of thousands of small probabilities underflows; a sum of logs does not.

Differentiable sums

Sums differentiate term by term — gradient methods apply directly.

The i.i.d. assumption is doing real work here. Time series, spatial data and grouped data violate it, and the factorisation is then simply wrong.

Example: A Biased Coin

n flips, h heads, parameter $\theta = P(\text{heads})$:

$$\ell(\theta) = h \log \theta + (n - h) \log(1 - \theta) \quad \frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n - h}{1 - \theta} = 0 \quad \hat{\theta}_{\text{ML}} = \frac{h}{n}$$

The estimator is the observed frequency — the answer intuition would have given. MLE **derives** it rather than assuming it, and the same derivation works where intuition has nothing to offer.

Mini-Exercise

You flip a coin **3 times** and see 3 heads.

1. What is $\hat{\theta}_{\text{ML}}$?
2. Is that a reasonable belief about the coin?
3. What is the estimator missing?

Mini-Exercise — Solution

Maximum likelihood picks the parameters under which the observed data is most probable. It is the bridge from a **model of the world** to a **learning algorithm** — and it is blind to prior knowledge and to its own uncertainty.

Mini-Exercise — Solution

1. $\hat{\theta} = 3/3 = 1$ — the coin never comes up tails.

2. No. Three flips is no basis for excluding tails entirely.

3. Any **prior** knowledge that coins are usually near-fair, and any account of its own **uncertainty**. MLE returns a single point and no error bars.

Both gaps are what the Bayesian treatment later repairs.

Maximum likelihood picks the parameters under which the observed data is most probable. It is the bridge from a **model of the world** to a **learning algorithm** — and it is blind to prior knowledge and to its own uncertainty.

The Same Model, Told Differently

Classical account

Fit a line by **minimising squared error**:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i (y_i - \mathbf{w}^\top \mathbf{x}_i)^2$$

Justified after the fact by Gauss–Markov: it is the best linear unbiased estimator.

Probabilistic account

State how the data **arose**:

$$y = \mathbf{w}^\top \mathbf{x} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Then estimate $\mathbf{w}$ by maximum likelihood.

Do these give the same answer?

The Derivation

The noise model says $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i \mid \mathbf{w}^\top \mathbf{x}_i, \sigma^2)$, so

$$\begin{aligned} \ell(\mathbf{w}) &= \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \mathbf{w}^\top \mathbf{x}_i)^2}{2\sigma^2}\right) = \text{const} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{w}^\top \mathbf{x}_i)^2 \end{aligned}$$

Maximising the log-likelihood **is** minimising the residual sum of squares. The constant and the $1/2\sigma^2$ do not move the arg max. $\hat{\mathbf{w}}_{\text{ML}} = \hat{\mathbf{w}}_{\text{OLS}}$

What the Probabilistic View Adds

The assumptions become visible

Additive noise, **zero mean**, **constant variance** (homoscedasticity), **independent** across observations, **Gaussian**.

They become checkable

Plot the residuals. Fanning out? Variance is not constant. Skewed? Not Gaussian. Patterned? Not independent.

You get a distribution

Not just $\hat{y}$ but $p(y \mid \mathbf{x})$ — a **predictive distribution** with an interval, not a point.

Choosing a loss and choosing a noise model are the **same decision**. Squared error $\Leftrightarrow$ Gaussian noise absolute error $\Leftrightarrow$ Laplace noise, which is why it is more robust to outliers.

Mini-Exercise

You fit a linear model to house prices. The residual plot shows the spread of errors **growing** with the predicted price.

1. Which assumption is violated?
2. Is $\hat{\mathbf{w}}$ still meaningful?
3. What is definitely wrong?

Mini-Exercise — Solution

Least squares **is** maximum likelihood under Gaussian noise. Stating the model probabilistically changes no estimate — it exposes the assumptions and buys you a predictive distribution.

Mini-Exercise — Solution

1. Homoscedasticity — the noise variance is not constant.

2. The point estimate remains unbiased it is just no longer the minimum-variance one.

3. Every **interval** and every **p-value**: they are computed from a single σ that does not exist. Predictions are over-confident for expensive houses and under-confident for cheap ones.

Least squares **is** maximum likelihood under Gaussian noise. Stating the model probabilistically changes no estimate — it exposes the assumptions and buys you a predictive distribution.

Module 3

From Regression to Classification

***Guiding Question:** A decision is a yes or a no. What goes wrong when we fit a straight line to one?*

Fitting a Line to Labels

Encode the two classes as $y \in \{-1, +1\}$, fit a linear model, and classify by the sign of the prediction:

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x})$$

One feature

The **discrimination point** x' satisfies

$$w_0 + w_1 x' = 0$$

Everything to one side is one class.

Two features

The **discrimination line** satisfies

$$w_0 + w_1 x_1 + w_2 x_2 = 0$$

In p dimensions it becomes a hyperplane.

Where It Breaks

Squared loss punishes correctness

A point far on the *correct* side has a large residual, so least squares drags the boundary toward it.

The boundary shifts

Adding well-classified points — or an outlier — moves the discrimination point, even when nothing was misclassified.

The output is also unbounded: $\mathbf{w}^\top \mathbf{x}$ can be 17 or -40 , and neither is interpretable as a degree of belief.

Two Problems, One Fix

$\text{sign}(\mathbf{w}^\top \mathbf{x})$ is not differentiable

No gradient means no gradient descent — the optimisation machinery does not apply.

The boundary is unstable

Squared loss on ± 1 labels lets well-classified points shift the discrimination point.

Replace the hard threshold with the **logistic sigmoid**:

$$h_{\Theta}(\mathbf{x}) = \text{expit}(\mathbf{w}^\top \mathbf{x}) \approx P(y = 1 \mid \mathbf{x})$$

The output becomes a smooth, bounded quantity that can be read as a probability — and can be differentiated.

Odds and Log-Odds

Odds

The ratio of favourable to remaining cases.

Rolling a six: odds 1 : 5.

Logit

The **log-odds** of a probability p :

$$\text{logit}(p) = \ln \frac{p}{1 - p}$$

Range

$p \in (0, 1)$ maps to $(-\infty, +\infty)$
— exactly the range a linear model produces.

The modelling assumption: the logit of the target is a linear combination of the features.

$$\ln \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}$$

The Model

Inverting the logit gives the **logistic sigmoid**:

$$h_{\Theta}(\mathbf{x}) = \text{expit}(\mathbf{w}^{\top} \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^{\top} \mathbf{x}}}$$

Shape

An S-curve: near-linear around 0, saturating towards 0 and 1 at the extremes.

Reading

$h_{\Theta}(\mathbf{x}) = P(y = 1 \mid \mathbf{x})$, and the decision boundary is still $\mathbf{w}^{\top} \mathbf{x} = 0$ — where the sigmoid passes 0.5.

The Loss: Cross-Entropy

Binary cross-entropy (log loss)
$$L(\Theta) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \ln h_{\Theta}(\mathbf{x}_i) + (1 - y_i) \ln (1 - h_{\Theta}(\mathbf{x}_i)) \right]$$

For $|Y| > 2$ classes, the multiclass form sums over the classes with a softmax output.

Where does it come from? From the **likelihood**: $P(y_i = 1 \mid \mathbf{x}_i, \Theta) = h_{\Theta}(\mathbf{x}_i)$ and $P(y_i = 0 \mid \mathbf{x}_i, \Theta) = 1 - h_{\Theta}(\mathbf{x}_i)$. Maximising the likelihood of the data is minimising this loss.

Why It Fixes the Problem

Confident and correct

Loss goes to zero — no penalty for being emphatically right.

Confident and wrong

Loss goes to infinity — exactly the errors we want punished.

There is no closed-form solution the parameters are found by gradient descent on this loss.

Takeaway

Logistic regression is a **linear model of the log-odds**, trained by maximising likelihood. It keeps the linear decision boundary but produces a calibrated probability — and a loss that punishes the right kind of error.

Module 4

Logistic Regression, Probabilistically

*Guiding Question: Cross-entropy was chosen because it punishes confident mistakes.
Where does it actually come from?*

The Same Pattern Again

Classical account

Squash a linear score with a sigmoid, then minimise **cross-entropy** because it punishes confident mistakes.

Probabilistic account

State the generative story: the label is a **Bernoulli** draw,

$$y \mid \mathbf{x} \sim \text{Bernoulli}(\sigma(\mathbf{w}^\top \mathbf{x}))$$

then maximise the likelihood.

The Derivation

With $h_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, one observation has probability $h_i^{y_i} (1 - h_i)^{1-y_i}$, so

$$\ell(\mathbf{w}) = \sum_{i=1}^n \left[y_i \log h_i + (1 - y_i) \log(1 - h_i) \right] = -n \cdot (\text{cross-entropy})$$

Cross-entropy **is** the negative log-likelihood of a Bernoulli model. As with squared error and the Gaussian, the loss was never an independent choice.

Gaussian noise

→ squared error → linear regression

Bernoulli labels

→ cross-entropy → logistic regression

What It Adds: Calibration

The output is now a **claim about frequencies**: among all cases predicted at 0.7, roughly 70% should be positive.

Calibrated

A reliability plot — predicted probability against observed frequency — lies on the diagonal. The number can be fed into a cost-based decision.

Miscalibrated

Systematically over- or under-confident. Accuracy can still be high the probabilities are simply not usable as probabilities.

Accuracy does not test calibration and calibration does not test accuracy. A model that outputs the base rate for everyone is perfectly calibrated and useless.

Regularisation Is a Prior

Adding an L_2 penalty to the loss:

$$\arg\max_{\mathbf{w}} \underbrace{\log p(D \mid \mathbf{w})}_{\text{likelihood}}$$

- $\underbrace{\log p(\mathbf{w})}_{\text{prior}} \propto p(\mathbf{w}) = \mathcal{N}(0, \tau^2 \mathbf{I})$

L_2 regularisation is exactly a **zero-mean Gaussian prior** on the weights, and the regularised estimate is the **MAP** estimate. L_1 corresponds to a Laplace prior, whose peak at zero is why it produces sparsity.

Cross-entropy is the Bernoulli negative log-likelihood **regularisation is a prior** and the probabilistic reading turns two heuristics into one principle — maximise likelihood, optionally times a prior.